

## 4. Exponentials and logarithms

### What students need to learn:

$y = a^x$  and its graph.

Laws of logarithms.

To include

$$\log_a xy \equiv \log_a x + \log_a y,$$

$$\log_a \frac{x}{y} \equiv \log_a x - \log_a y,$$

$$\log_a x^k \equiv k \log_a x,$$

$$\log_a \frac{1}{x} \equiv -\log_a x,$$

$$\log_a a = 1$$

The solution of equations of the form  
 $a^x = b$ .

Students may use the change of base formula.

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